WBR Perception — Progress Report

Process, results and next steps

Intel RealSense D435i

Raspberry Pi 5 / ROS 2 Jazzy

ORB-SLAM3 stereo-inertial

STM32H7 controller boundary



D435i hardware illustration

Owner's operating guide 13 October 2026

Where we are

MEASURED / OPEN

Today

A native ARM64 build produced three usable EuRoC trajectories. D435i recorder and estimator binaries are built; neither has run on this camera.

Next

Repair power, preserve/migrate the OS card, qualify data storage, then re-establish N1 → N2 → N3 on the new unit.

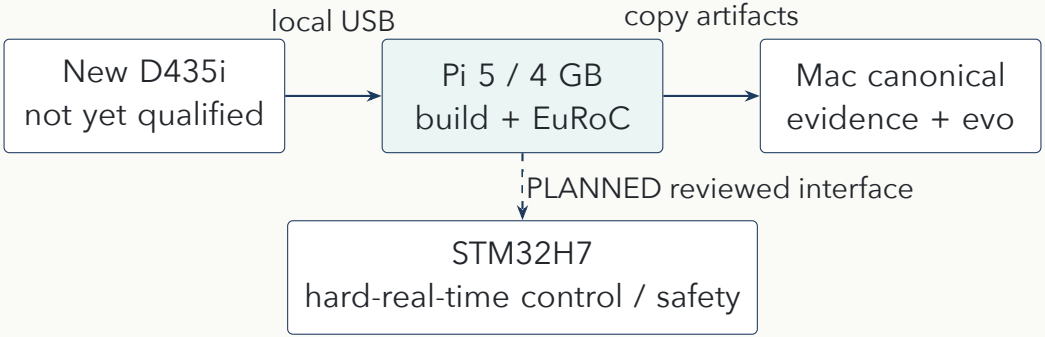
Reading route

Where we are	pp. 2–5	Today's work	6–19
Two programs	20–28	Runbook	29–39
Tools and sources	40–42	Agent tickets	43–44

All gates remain OPEN. Every command in the runbook is a future instruction.

1 / Whole system: today is on the perception side

SOURCE-SUPPORTED / MEASURED / PLANNED



Ownership

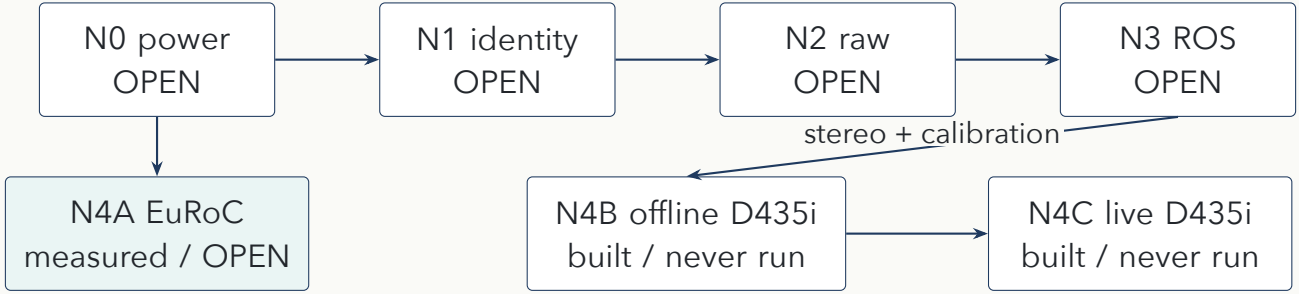
Acquire locally on the Pi. SSH controls the session; it is not a sensor data path. Keep Mac and Pi checkouts intact.

Current boundary

The EuRoC result exercises perception only. It establishes no robot integration, motor safety or controller interface.

1 / Gate ladder: evidence does not transfer sideways

OPEN / MEASURED / PLANNED



N4A tests known inputs independently of camera gates. N4B needs the physical gates plus a reviewed stereo extension and calibration. N4C adds live scheduling.

N5 profiling, N6 health and N7 controller integration are still PLANNED.

1 / What changed: treat this as a new unit

MEASURED / OPEN

Field	Observed 13 October
Pi / boot storage	Pi 5, 4 GB; Ubuntu 24.04.4; 32 GB microSD, no NVMe
SDK device serial	`317422076650`: recorder, ROS serial_no and gate selection
ASIC serial	`327743061057`: separate device identity field
Firmware-update ID	`327743061057`: separate observed field
USB descriptor serial	`327743061057`: udev / IIO matching
Firmware / link	5.17.3.10; reported BMI085; USB 3.2 at 5000 Mbps

Old SDK `243322073847`, USB `251843061466`: September transport/calibration receipts do not qualify this unit.

> Fresh N1 must read every identity field; an equal value does not make fields interchangeable.

2 / Platform and power: what actually ran

MEASURED / INFERRED

Measured platform	Measured warning
Pi 5 / 4 GB, kernel 6.8.0-1064-raspi, ROS 2 Jazzy binary, 5 V / 3 A supply. Temporary 4 GB swap helped the build complete.	Live under-voltage appeared during load: 46 events during the build; 87 of 178 five-second samples during SLAM runs, or 49% rounded.

A build succeeding under poor power is evidence of compilation, not an eligible performance platform.

```
vcgencmd get_throttled
vcgencmd measure_temp
timedatectl status
```

> Fix the supply first; then record fresh health under the same intended load.

2 / Decode 0x50005: live flags versus history

MEASURED / SOURCE-SUPPORTED

Bit / mask	Meaning in the runbook
0 / 0x1	Under-voltage is occurring now
2 / 0x4	Throttling is occurring now
16 / 0x10000	Under-voltage has occurred since boot
18 / 0x40000	Throttling has occurred since boot

Thus `0x50005` contains both live faults and latched history.

Strict qualification

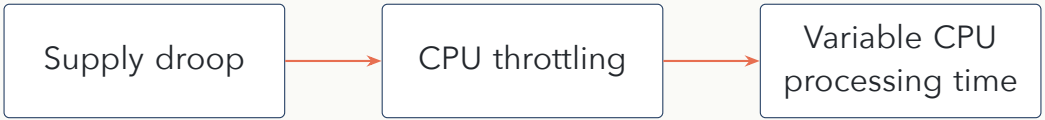
Require `0x0`. Preserve old power logs. After correcting the cause, an owner-chosen reboot starts a fresh observation interval.

Diagnostic exception

Explicit latched-only diagnosis can admit `0x50000`; live bits still stop. Result is OPEN_DIAGNOSTIC and cannot enter accepted N3.

2 / Why timing is unusable, while accuracy is informative

MEASURED / INFERRED / OPEN



Observed duration

184 s dataset, 282–301 s wall time. Wall time includes vocabulary load, initialization and shutdown; it is not a per-frame tracking benchmark.

What survives

Final trajectory error remains measurable against the dataset reference. It supports this algorithm path on MH01, under this recorded run context.

No real-time claim follows. Re-measure on stable power; separate startup, per-frame work, queue residence and shutdown. The target 8 GB Pi also needs its own measurements.

2 / Native build: what the build proved

MEASURED / OPEN



Build evidence

Pangolin v0.9.1 built in 178 s. Main library compilation took about 71 min at `-j2`. Preserve build scripts and Pi-side logs/hashes.

Meaning

A usable binary exists for this compiler and dependency set. EuRoC execution is the separate behavioral check; compilation alone cannot establish estimator correctness.

Upstream acquisition example was unchanged for N4A; the library had three documented compatibility patches.

2 / Build patch 1: language standard and CPU flags

MEASURED / SOURCE-SUPPORTED

Before	After
C++11 build settings and `-march=native` belonged to upstream build assumptions. New compiler/dependency combinations need an explicit compatibility configuration.	Use C++17 and remove `-march=native`. Keep the resulting compiler flags in provenance rather than assuming the Mac and Pi build the same code.

This is a build configuration change, not a change to stereo features, inertial equations or trajectory evaluation.

```
# Read-only inspection of the preserved native build
cat build/CMakeFiles/ORB_SLAM3.dir/flags.make
```

> Record the exact flags and architecture; a successful build does not prove portable binaries.

2 / Build patches 2 and 3: two concrete failures

MEASURED / SOURCE-SUPPORTED

Patch	Failure and bounded fix
Sophus tests OFF	Sophus unit tests failed with GCC 13 `-Werror=array-bounds` on aarch64. ORB-SLAM3 uses Sophus headers; skip these third-party tests, retain the failure log.
`bool++` to true	C++17 forbids incrementing a bool. Replace `mnFullBAIdx++` with `mnFullBAIdx = true`; this preserves the old bool increment result.

The disabled Sophus tests are a validation limitation. They do not constitute a passing third-party test suite.

> Keep patches attached to source identity and build evidence; do not edit canonical upstream to hide the failure.

2 / Swap: a build aid, not performance evidence

MEASURED / INFERRED

Observed	Inference to test
4 GB RAM plus a temporary 4 GB swapfile. Free RAM fell to about 130 MB during compilation. This is why `-j2` and swap mattered.	An 8 GB Pi may build without swap. That is a hypothesis; no 8 GB build result is present in today's evidence.

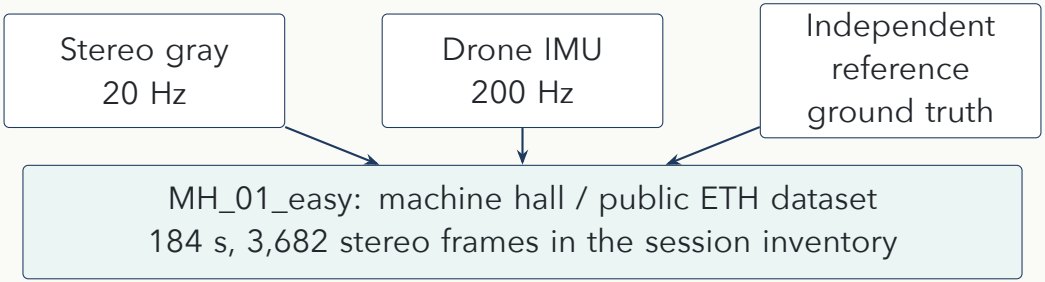
```
free -h
swapon --show
df -B1 /
```

Build pressure and runtime pressure are different workloads. N4A peak RSS was 823–828 MiB; neither that figure nor the swapfile size is the total system memory requirement.

> Log RAM, swap use and storage headroom; avoid extra flash writes during capture.

2 / EuRoC MH01: known input before camera adaptation

MEASURED / SOURCE-SUPPORTED



The dataset supplies images, motion samples, timestamps and its own settings. Our D435i is absent from this test, so its USB link and calibration cannot cause these input failures.

First establish that the native estimator can use known, calibrated input.

2 / Ground truth: an independently measured answer

SOURCE-SUPPORTED / MEASURED

Ground truth is the reference pose trajectory used to score the estimate. It is obtained with external measurement equipment, not by trusting ORB-SLAM3's own map.

EuRoC reference

Machine-hall data uses laser-tracker position measurements; the dataset provides processed state ground truth. Vicon-room sequences use motion capture. Do not claim MH01 has Vicon measurements.

Evaluation here

Use `state_groundtruth_estimate0/data.csv`, convert to TUM, and compare body/IMU poses in the same frame convention and time units.

A hand-carried D435i take has no external ground truth in the current plan. A closed loop or a plausible path cannot substitute for one.

> Ground truth still has coverage, timing and measurement uncertainty to document.

2 / ATE: remove the arbitrary origin, keep scale

SOURCE-SUPPORTED / MEASURED method

Let $\hat{p}_i$ be an estimated body position and p_i its time-associated reference position. Fit one rigid transform to the whole accepted trajectory:

$$(R^*, t^*) = \arg \min_{R \in SO(3), t} \sum_{i=1}^{n} \|R\hat{p}_i + t - p_i\|^2.$$

Then $r_i = R^*\hat{p}_i + t^* - p_i$ and

$$\mathrm{ATE}_{\mathrm{RMSE}} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} \|r_i\|^2}.$$

SE(3) means rotation + translation only. Stereo-inertial has metric scale; adding scale fitting (Sim(3)) would conceal scale error.

> Alignment removes coordinate choice; the remaining residuals measure position mismatch.

2 / ATE alignment: schematic before and after

SOURCE-SUPPORTED / ASSUMED schematic



No axes or numerical errors are implied by this drawing. ATE uses all accepted matched positions, not just the four illustrated points.

> No scale change is permitted; residual lengths are combined by RMS.

2 / evo: reproduce the score and distinguish RPE

MEASURED ATE protocol / CALCULATED RPE

Run	RPE trans. RMSE (m)	RPE rot. RMSE (deg)
1	0.013295	0.076464
2	0.013559	0.076776
3	0.014021	0.074883

SE(3); shared EuRoC clock assumed, offset 0 s; association tolerance 0.02 s (3,639 matched pose pairs per run). One endpoint per eligible start: first later reference pose ≥ 1 s later; no interpolation. These are `traj_eval.py` results; its pair policy differs from evo's RPE filters.

> About 1.3 cm RPE over 1 s describes local drift; ATE scores the aligned global path.

2 / Three runs: score range, not a single best screenshot

MEASURED / SOURCE-SUPPORTED / CALCULATED

Run	ATE RMSE	ATE max	Wall time	Peak RSS
1	0.0415 m	0.081 m	301 s	823 MiB
2	0.0368 m	0.076 m	282 s	828 MiB
3	0.0416 m	0.088 m	284 s	828 MiB

All three: one map, no resets, IMU initialized, exit 0; 3,678 output poses. Mean RMSE about 0.040 m; range 0.0368–0.0416 m. **Verified paper comparator: 0.036 m.** Campos et al., Table II, MH01 stereo-inertial; median of ten runs, processed GT, SE(3). Our three-run mean is a different statistic and protocol equivalence is not fully established.

> Similar error magnitude is useful baseline evidence, not a formal reproduction or gate acceptance.

2 / Variation and gate status: what N4A establishes

MEASURED / INFERRED / OPEN

Supported by these trials	Still unproven
The native ARM64 estimator and compatibility fixes produce finite, initialized trajectories on MH01 with centimeter-scale ATE near the published comparator.	D435i geometry/noise/clock, physical transport, other sequences and healthy real-time performance. N4A RPE is now CALCULATED (p. 17). Numeric acceptance budgets remain OPEN.

Mapping and loop-closing threads can execute in different orders. Identical input need not produce identical keyframes; observed counts were 126 / 126 / 129. Scheduling is a plausible explanation, not a measured cause of the score variation. Pi UTC labels were about 80 min behind before NTP was restored near 09:45Z. Retain the offset note; do not align Mac/Pi logs by uncorrected labels.

N4A remains OPEN; power and protocol/budget review are unresolved.

3 / Two programs: record once, run the estimator later

MEASURED build / OPEN execution

`d435i_record`

An acquisition utility. Uses librealsense directly, records left/right IR and raw gyro/accel, and writes a timestamp audit. It does not run SLAM.

`stereo_inertial_realsense_D435i`

A patched upstream input example. Feeds stereo images and associated IMU samples to ORB-SLAM3, then saves body and keyframe trajectories.

Built bundle: `~/wbr-runtime/n4b-20261013T080913Z/bin/`. Linkage was reported clean. Neither binary has been exercised with this camera or a real recording.

> Run one camera-owning process at a time. A clean link is not a validated input path.

3 / Recorder profile: stereo development adds bandwidth

SOURCE-SUPPORTED / PLANNED acquisition

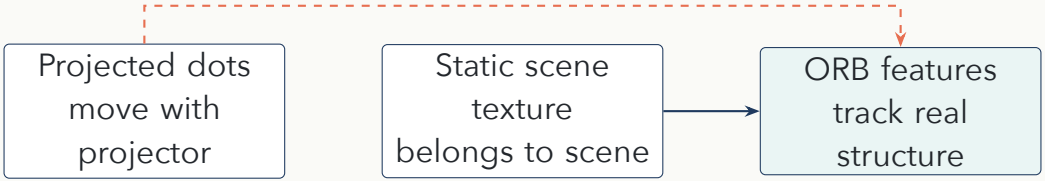
Stream	Explicit selected profile
IR1 and IR2	Left/right infrared, each 640×480, Y8, 30 Hz
Gyroscope	Raw XYZ angular velocity, 200 Hz
Accelerometer	Raw XYZ acceleration, 100 Hz
Excluded	No depth/RGB; no rate fallback; no combined ROS IMU

This is the N4 development profile. Strict N2 stays IR1-only at 15 Hz with gyro/200 and accel/100. N3 preserves that baseline before the reviewed stereo extension.

> Selecting a supported profile does not measure actual delivery under recording load.

3 / Emitter OFF: features must belong to the scene

SOURCE-SUPPORTED / PLANNED readback



Projected texture has view-dependent behavior and can confuse feature identity between cameras and over time. Both live programs set the emitter OFF and read the option back before streaming.

Prepare scene

Diffuse light, textured static objects, short exposure and modest motion; inspect IR for blur and saturation.

Preserve context

Record exposure metadata and motion-correction/global-time settings. Emitter OFF alone does not qualify image quality or timestamps.

3 / Recorder outputs: timestamps are evidence, not truth

SOURCE-SUPPORTED / OPEN

One new take directory

`stereo_imu.db3`: SDK recording.
`timestamps.csv`: full-precision stream/time/IMU audit.
`sdk.log`: unfiltered SDK messages.

What to reconcile

Stored decoded counts versus callbacks; image pixels and strides; timestamp domains, frame steps, boundaries and left/right pairing. Count differences must be explained.

The recorder uses device timestamps. Host arrival/steady time serves a separate audit purpose; it cannot silently replace sensor time in SLAM. Local source now names `.db3`; older preparation prose still says `.bag`. Use the built-binary/source hash and actual finalized output. Confirm SDK playback compatibility rather than assuming an ordinary ROS topic bag.

> Normal stop finalizes the file. Hash it before playback and keep every failed take.

3 / Storage arithmetic: 18.4 MB/s is only the floor

CALCULATED / MEASURED history / OPEN

$$2 \times 640 \times 480 \times 1 \times 30 = 18{,}432{,}000\ \mathrm{B/s} = 18.432\ \mathrm{MB/s}.$$

Pixels alone: 1.106 GB/min; a 90 s take is 1.659 GB. Raw motion floats add 216 kB/min before timestamps, metadata, serialization and CSV.

Historical warning

September card probe about 19 MB/s, worst about 17 MB/s: too little margin against this stereo payload. Today's free space must be rechecked after build/swap.

Planning allowance

Preparation proposes 1.5 GB/min provisionally. Actual finalized size, sustained/peak writes and backlog are still UNKNOWN. Card labels are not measurement evidence.

> GB-scale bags require separately qualified storage. The larger OS card does not satisfy that prerequisite by capacity alone.

3 / Patched estimator: one executable, two modes

SOURCE-SUPPORTED / MEASURED build / OPEN

```
# N4B: immutable recording, playback is not real-time
$WBR_SLAM VOCAB YAML run01 --bag TAKE.db3 --no-viewer
# N4C: same vocabulary and YAML, direct live source
$WBR_SLAM VOCAB YAML live01 --no-viewer
```

`--bag FILE`

File-only device resolution, no loop, preserved source stamps. Disable real-time playback before start; allow processing to finish without dropping to wall-clock pacing.

`--no-viewer`

Disable the GUI. Pangolin remains a linked dependency; headless behavior does not eliminate its build requirement.

> These are future command shapes. N4B isolates input/calibration; N4C adds live overload and source-age questions.

3 / IMU patch: interpolate within valid brackets

SOURCE-SUPPORTED / synthetic-tested / OPEN SDK behavior

$$a(t_g) = (1-\alpha)a_0 + \alpha a_1, \quad \alpha = (t_g - t_0)/(t_1 - t_0)$$

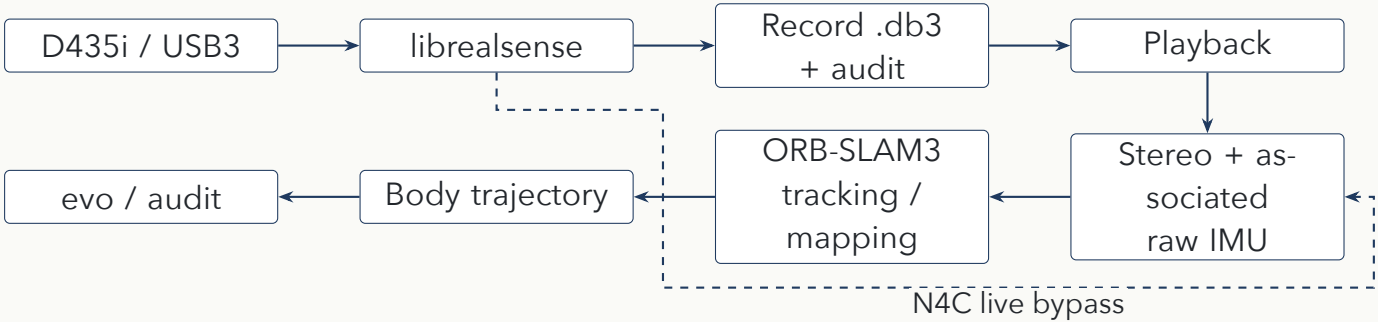


Fix upstream interpolation overwrite; send each gyro once with bracketed accel. Retain the endpoint at/after an image for preintegration. Reject nonfinite, duplicate/backward samples and interior motion gaps over 50 ms; never extrapolate. Explicitly count uncovered startup/EOF images and samples. A gyro-grid input at 200 Hz still contains measured accel at 100 Hz.

Synthetic tests verify association logic only; real-file EOF/callback behavior remains OPEN.

3 / End-to-end path: offline before live

SOURCE-SUPPORTED / PLANNED execution



Frozen data enables repeated diagnosis without repeating hand motion. The eight-pair image queue owns pixels; playback waits for space, while live overload fails explicitly. Neither mode permits silent latest-frame overwrite.

> Repeated offline runs can still vary because mapping is threaded. Preserve all outputs.

3 / Configuration register: readback versus calibration

MEASURED readback / ASSUMED noise / OPEN

Quantity	Candidate YAML / confidence
IR intrinsics	$f_x = f_y = 386.95$, $c_x = 321.03$, $c_y = 241.04$ px at 640×480; SDK rectified readback
Stereo baseline	$b = 0.0499126$ m; positive magnitude of IR1→IR2 translation
`IMU.T_b_c1`	Camera→gyro/body, identity rotation; $t = (-5.52, +5.10, +11.74)$ mm; nominal factory geometry
IMU noise / walks	$10^{-3}, 10^{-2}, 10^{-6}, 10^{-4}$ in upstream key order; ASSUMED, not unit measurements

Keep the reported "IMU Calibration is not available" warning. Fresh SDK logs must establish its occurrence on this unit; old warning logs are not new-unit evidence.

> Noise, bias/scale, actual alignment and camera–IMU time offset remain unqualified. Built binaries have never run here.

4 / Runbook entry: prepare a bounded session

PLANNED / OPEN

Before any execution

User must explicitly say CAMERA AVAILABLE before camera-dependent commands or ORB execution. Capture locally on Pi; use a verified runtime bundle, fresh evidence directory and exact active-unit selection.

Order of decisions

Power → OS migration → separate data storage → N1 → N2 → N3 → stereo development take → N4B → calibration → qualified offline review → N4C → profiling.

```
# Alias declarations; resolve actual bundle paths first
export WBR_BUNDLE="VERIFIED_RUNTIME_BUNDLE"
export WBR_SLAM=$WBR_BUNDLE/bin/stereo_inertial_realsense_D435i
export WBR_DATA="QUALIFIED_SEPARATE_DATA_MOUNT"
```

> The next pages specify acceptance conditions; they do not report PASS. Stop on nonzero exit and inspect evidence before progressing.

4 / Step 1: repair power and establish N0 eligibility

PLANNED / OPEN

Goal / tool: stable supply and cooling; `vcgencmd`, inventory and UTC service. Ordered supply: 5.1 V / 5 A; verify the physical label/cable.

```
python3 platform_inspect.py > NEW-inventory.json
vcgencmd get_throttled
vcgencmd measure_temp
timedatectl status
```

Expected / acceptance

ARM64 Ubuntu 24.04, Jazzy, synchronized UTC, flags `0x0`, temperature $< 75\,^{\circ}\mathrm{C}$, MemAvailable ≥ 512 MiB. Repeat health under intended load.

Failure → inspect

Live voltage bits → supply/cable/connectors. Heat → cooling. Latched bits → retain history, correct cause, owner-chosen reboot then fresh evidence; do not mask flags.

> The supply rating is a plan. Fresh healthy observations determine eligibility.

4 / Step 2: migrate the OS card, qualify data separately

PLANNED / OPEN / CALCULATED capacity

Goal / tool: image the 32 GB card on Mac, write the verified clone to 128 GB V30/A2, grow root; keep the original card as rollback.

```
diskutil list   # Mac: inspect disks before a reviewed clone
lsblk -f; findmnt -T /; df -B1 /   # Pi after migration
```

Expected / acceptance	Failure → inspect
Image hash and readback; original retained; clone boots with files/versions intact, root size checked and fresh N0.	Wrong root size → growth procedure; boot/files missing → image/hash and disk identity. Retain the original.

About 95 GB data capacity is arithmetic only. Large bags require a separate qualified mount: `storage_qualify.py` with inspected UUID/device, then fresh N1. See [G] for the full storage contract.

4 / Step 3: fresh N1 on the new D435i

PLANNED / OPEN

Goal / tool: verify camera's own USB branch, normal-user permissions, device identity and healthy platform; `n1_precheck.py`.

```
python3 n1_precheck.py --camera-available \
  --direct-cable-confirmed --output "$WBR_RUN/n1"
```

Expected / acceptance

Sealed `result.json`, own branch ≥ 5000M, direct cable, video/HID/IIO access, all four identity fields, firmware readback; strict health. Receipt matches host/boot and is at most two hours old.

Failure → inspect

480M → cable/port/hub, stop before SDK. Motion permission denied → serial-matched HID/IIO controls. Wrong serial → SDK versus descriptor selection. No broad permissions or firmware update.

> N1 previously stopped at the clock prerequisite. NTP repair alone does not produce a new N1 receipt.

4 / Step 4: N2 strict raw transport and mandatory soak

PLANNED / OPEN

Goal / tool: prove native delivery before ROS/SLAM; `phase_gate.py`. IR1 Y8 640×480/15, gyro/200, accel/100, unchanged.

```
python3 phase_gate.py n2 --camera-available --seconds 20 \
  --n1-result "$WBR_RUN/n1/result.json" --output "$WBR_RUN/smoke"
# Then 105 s, then 300 s; chain --previous sealed receipts
```

Expected / acceptance

Raw CSV/logs, health and images. IR 14.25–15.75 Hz; gyro 190–210; accel 95–105. Image gap ≤ 0.20 s, motion ≤ 0.05 s; positive stamp/frame steps, no missing image numbers.

Failure → inspect

Rate/gap → CSV boundaries; USB reset → journal; voltage → supply. Review images and warning; `accept-n2` requires actual review.json.

The 300 s raw soak is mandatory. A 105 s run or OPEN_DIAGNOSTIC cannot replace acceptance.

4 / Step 5: N3 ROS record and isolated replay

PLANNED / OPEN wrapper state

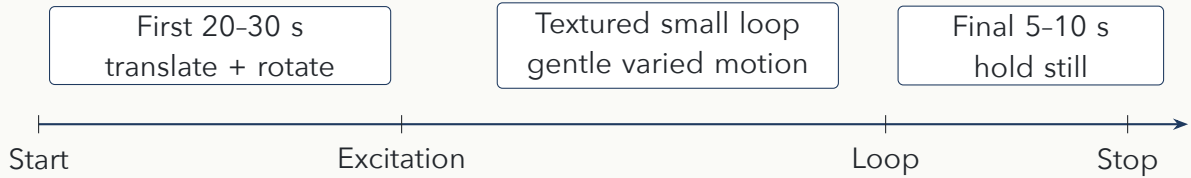
Goal / tools: qualify ROS delivery/record/replay with realsense-ros, rosbag2 and observers. Verify installed wrapper; 4.58.4 is only the prepared target.

```
python3 n3_record.py --camera-available --n1-result N1.json \
  --n2-result N2.json --storage-result STORAGE.json \
  --bag QUALIFIED/NEW_BAG --seconds 20 --output RECORD
python3 n3_bag.py replay QUALIFIED/NEW_BAG \
  --record-result RECORD/result.json --live-domain 42 \
  --replay-domain 43 --output-dir REPLAY
```

Expected / acceptance: 60 s live observer, N2 rate/gap bands under load; exact topics/types/frames/context and finalized metadata. Stop live source; replay counts and decoded payload hashes must match. **Failure → inspect:** context/QoS → version/readback; replay contamination → processes/domains. Review stereo separately.

4 / Step 6: first guided stereo take, 60–90 seconds

PLANNED development / OPEN qualification



Recipe: translate 10–30 cm in all axes with ± 15–$30°$ rotations, keep texture visible, avoid shocks/blur. The timing bars are approximate recipe stages, not a measured trajectory.

```
# Recorder creates TAKE exclusively; parent is on qualified storage
$WBR_BUNDLE/bin/d435i_record "$WBR_TAKE" 90
```

Expected / acceptance: all four callback streams, finalized .db3 + CSV + raw SDK logs, emitter readback OFF; decoded counts/pixels/stamps reconciled and actual 30 Hz stereo assessed. “OPEN: recorded” is not acceptance. Missing streams/error → raw logs/profile/storage.

4 / Step 7: offline N4B run, then evaluate honestly

PLANNED / OPEN / no external GT for hand take

Goal / tool: diagnose the fixed take before live; patched example plus trajectory/timestamp audit.

```
$WBR_SLAM VOCAB YAML run01 \
  --bag "$WBR_TAKE/stereo_imu.db3" --no-viewer
```

Expected / acceptance	Failure → inspect
"IMU initialized"; finite ordered poses, map/reset timeline and input/rejection audit. One map/no resets is a development goal; numeric budgets remain OPEN.	"Not enough motion... Reseting" → excitation/time/calibration. Coverage timeout → brackets; vertical disparity → rectification; empty poses → raw logs.

Repeat the hashed take/config. Without independent ground truth, report continuity/repeatability and closure discrepancy; ATE is unavailable.

4 / Step 8: calibrate camera–IMU geometry and time

PLANNED / SOURCE-SUPPORTED / OPEN tool integration

Goal / tool: estimate spatial transform and time offset with Kalibr; plan a fixed Aprilgrid and excite all axes under good light.

```
# Future compatible Kalibr environment, not an installed Pi command
kalibr_calibrate_imu_camera --bag CALIBRATION.bag \
  --cam camchain.yaml --imu imu.yaml --target aprilgrid.yaml
```

Expected / acceptance: report PDF, result text and camchain-imucam YAML; review residuals, coverage and holdout behavior. Verify transform direction/units and offset sign once. Approved residual budgets are still OPEN. **Failure → inspect:** poor fit → blur/target coverage/time jitter/IMU model before tuning. Kalibr expects a compatible ROS bag; SDK .db3 import and topic/model compatibility need a reviewed, lossless conversion path. No converter is claimed ready.

> Resolve IMU intrinsics/noise too. Then repeat offline N4B with frozen accepted calibration before any N4C qualification.

4 / Step 9: measure IMU noise with Allan deviation

PLANNED / ASSUMED current YAML / OPEN

Goal / tool: replace example noise densities and bias walks with unit evidence. Candidate: `allan_variance_ros`; import/build compatibility has not been qualified.

```
TOOL --help   # Pin and review the selected interface first
TOOL STATIONARY_RAW_IMU CONFIG OUTPUT
```

Expected / acceptance

Stationary full-precision six-axis data, stamps/units/temperature and no clipping/gaps; deviation curves + density/walk fits. Duration from required averaging times, normally hours.

Failure → inspect

Curve artifacts → stamps/gaps/motion; weak long-time fit → duration. Review units; avoid double discretization in ORB.

The hand take is too short for slow bias. Duration and fit acceptance need review. Zero SDK variance does not mean noiseless hardware.

4 / Step 10: live N4C, then profile N5

PLANNED / OPEN thresholds

Goal / tools: retain the accepted offline configuration while testing live scheduling; same example, gate observers and `vcgencmd`.

```
$WBR_SLAM VOCAB QUALIFIED.yaml live01 --no-viewer
vcgencmd get_throttled   # Alongside health sampling
```

Expected / acceptance

Finite tracking + state events, source/host epochs; pairing, drop/overflow, queues/source age and CPU/RSS/thermal/power logs. Accuracy/latency/loss budgets need approval.

Failure → inspect

Live-only overflow → queues; growing delay → backlog; loss in both modes → input/calibration. Under-voltage → stop and repair.

N5 separates startup, tracking, queues and shutdown; report p50/p95/max with clock definitions. Missing counters remain unavailable. Measure healthy 4 GB and target 8 GB separately.

5 / Tool map: platform, transport and recording

MEASURED components / PLANNED physical use / OPEN wrapper

Tool	Job / step	Status here
vcgencmd / journal	Power, temperature, kernel faults; N0–N5	Used in saved evidence
librealsense	SDK enumeration, raw device/file input; N1/N2/N4	SDK present; new path unrun
Gate utilities	Sealed N1/N2/N3 checks, health and replay	Offline tests; gates OPEN
realsense-ros	ROS topics, parameters, frame/context; N3	Installed state to verify
rosbag2	ROS bag record/info/play; N3	Future qualification
d435i_record	SDK stereo/motion .db3 + audit; first take	Built, never run
Storage qualifier	Sustained writes/hash readback; data entry	Separate mount PLANNED

SDK recording and N3 ROS recording are distinct paths; they have different inventories and qualification evidence.

5 / Tool map: estimation, scoring and calibration

MEASURED baseline / ASSUMED config / PLANNED calibration

Tool	Job / step	Status here
ORB-SLAM3	Features + inertial tracking/mapping; N4	EuRoC ran; D435i unrun
Pangolin	Upstream visualization/build dependency	v0.9.1 built; headless link
Patched D435i example	Shared offline/live association; N4B/C	Built; SDK behavior OPEN
evo / traj_eval	Global ATE / local RPE; N4A/N5	ATE used; RPE CALCULATED
Kalibr	Camera–IMU transform/time + reports	PLANNED integration
Allan-variance tool	Noise/walk from stationary IMU	PLANNED tool selection
Hashes / manifests	Bind source, config, input and results	Required every stage

Choose the next tool from the unresolved question: delivery, input association, calibration or scheduling.

5 / Source key: how to verify each claim

SOURCE-SUPPORTED / OPEN evidence limitations

Key	Repository source (all under experiments/ unless stated)
[F]	2026-10-13-SESSION_FACTS_FOR_SLIDES.md: authoritative session summary
[A]	2026-10-13-n4a-euroc-mh01/: report, trajectories, evaluation and power logs
[B]	2026-10-13-n4b-d435i-prep/: candidate YAML, recorder, patch, calibration notes and plan
[G]	2026-10-13-pi5-env-snapshot/bundles/: active-unit runtime bundle PHASE_A_RUNBOOK.md

Build scripts: orbslam3-pi-build/; raw build logs also on Pi. Camera identity: pi5-env-snapshot/ADDENDUM_CAMERA_SWAP.md.

PROJECT_STATE/NEXT_GATE are historical drafts. Use current-unit evidence; course pages supply methods, not today's status.

6 / Direct agents with one uncertainty per ticket

PLANNED / SOURCE-SUPPORTED working rules

Example: fresh N1 ticket	Required delivery
Question: does SDK 317422076650 have direct ≥ 5000M access under healthy power? Allowed: inspect verified bundle, run only N1 after CAMERA AVAILABLE. Stop: any live power fault, 480M, identity mismatch or permission fault.	Raw USB/SDK/kernel logs; separate serial fields; boot/UTC/supply/cooling; source/binary hashes; sealed result + exact command/exit status; one unresolved blocker. No gate promotion from summaries.

Bound files and actions: checksummed Mac utility copies in a new Pi bundle, no checkout reset, no firmware/network change, no camera/ORB execution before authorization. Preserve failed runs and calibration warnings.

> Ask for evidence that resolves a question, not “make SLAM work.” Owner reviews the acceptance contract.

6 / Owner checkpoint: how to keep direction

PLANNED / OPEN

Ticket	Evidence to request before choosing the next ticket
Power + storage	Healthy loaded flags; disk/image identity/hash; qualified separate mount, fresh inventory
N1 → N2 → N3	Current-unit sealed receipts; mandatory 300 s raw soak; actual reviewed logs; finalized record and isolated replay
Development N4B	Hashed .db3/config/build; decoded input audit; initialization/reset timeline; every repeated trajectory, including failure
Calibration → N4C	Raw calibration/noise data + residuals + approved budgets; repeat offline review first; live queue/source-age/health evidence

At each handoff: changes, reproduction command, measurements, unknowns and the next discriminating check.

> Next: inspect the new supply and migrated OS; qualify separate data storage and fresh physical gates.